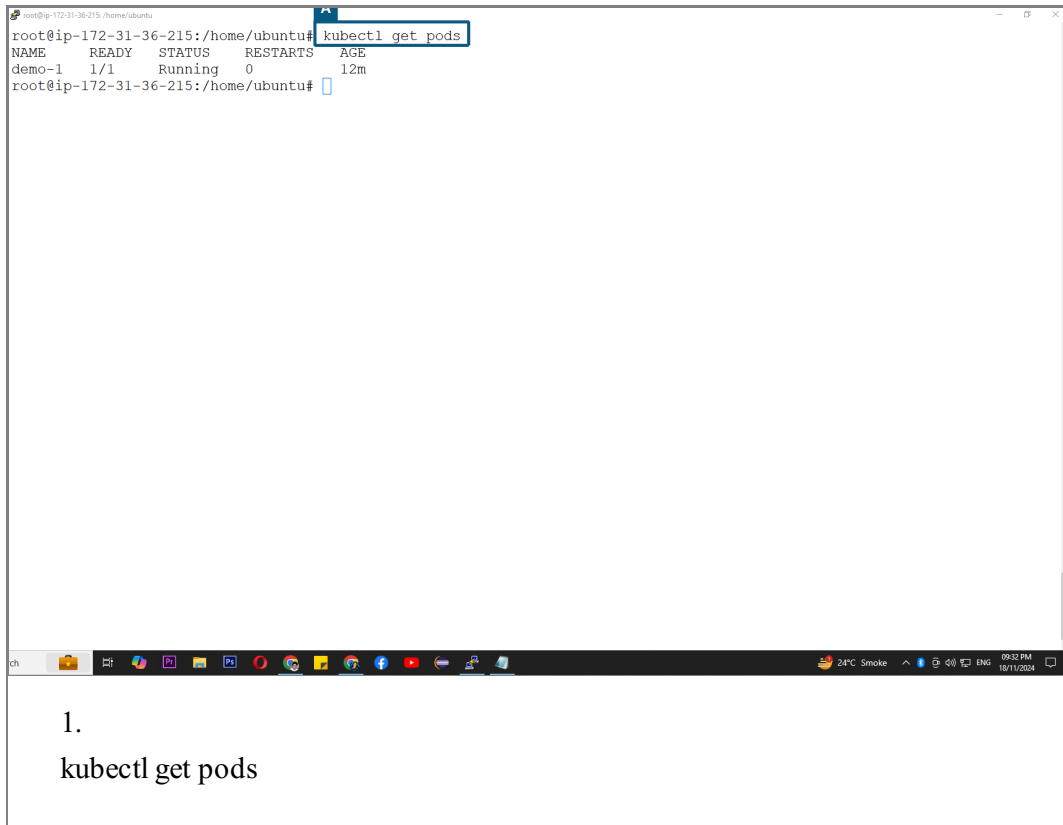

Basic commands in Kubernetes

Rohit Surwade

To get the pod's information



```
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
demo-1    1/1     Running   0           12m
root@ip-172-31-36-215:/home/ubuntu#
```

1.
kubectl get pods

Kubernetes Manifest to create a pod(with one container)

10,45 All

```
kind: Pod
apiVersion: v1
metadata:
name: testpod
spec:
containers:
- name: c00
image: ubuntu
command: ["/bin/bash", "-c", "while true; do echo Hello-Rohit; sleep 5; done"]
restartPolicy: Never # Defaults to Always
```

To Execute the Manifest

```
root@ip-172-31-36-215:/home/ubuntu#
A
B
root@ip-172-31-36-215:/home/ubuntu# kubectl apply -f manifest1.yml
pod/testpod created
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
demo-1    1/1     Running   0           15m
testpod    1/1     Running   0           20s
root@ip-172-31-36-215:/home/ubuntu#
```

3.

kubectl apply -f manifest_name.yml

To get extra information about the Pods

```
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods -o wide
NAME      READY   STATUS    RESTARTS   AGE   IP            NODE       NOMINATED NODE   READINESS GATES
demo-1    1/1     Running   0           18m   10.244.0.3    minikube   <none>            <none>
testpod    1/1     Running   0           3m42s 10.244.0.4    minikube   <none>            <none>
root@ip-172-31-36-215:/home/ubuntu#
```

4.

kubectl get pods -o wide

We can see the annotation's description in the pod

```
root@ip-172-31-36-215:/home/ubuntu# kubectl describe pod testpod
Service Account: default
Node: minikube/192.168.49.2
Start Time: Mon, 18 Nov 2024 16:26:45 +0000
Labels: <none>
Annotations: description: Creating a test POD
Status: Running
IP: 10.244.0.5
IPs:
  IP: 10.244.0.5
Containers:
  c00:
    Container ID: docker://0fe53e18eb299ab229dee0ec6019ef6217e832748fbbb20ab9407a7ded4a905
    Image: ubuntu
    Image ID: docker-pullable://ubuntu@sha256:278628f08d4979fb9af9ead44277dbc9c92c2465922310916ad0c46ec9999295
    Port: <none>
    Host Port: <none>
    Command:
      /bin/bash
      -c
      while true; do echo Hello-Rohit; sleep 5; done
    State: Running
    Started: Mon, 18 Nov 2024 16:26:48 +0000
    Ready: True
    Restart Count: 0
    Environment: <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-jpfqg (ro)
Conditions:
  Type              Status
  PodReadyToStartContainers  True
  Initialized         True
  Ready               True
  ContainersReady     True
  PodScheduled        True
Volumes:
  kube-api-access-jpfqg:
    Type: Projected (a volume that contains injected data from multiple sources)
```

8.

`kubectl describe pod pod_name`

To Delete a Pod

```
root@ip-172-31-36-215:/home/ubuntu# kubectl delete pod testpod
pod "testpod" deleted
```

9.

`kubectl delete pod pod_name`

Manifest to create Multi Container Pod

```
kind: Pod
apiVersion: v1
metadata:
  name: testpod3
spec:
  containers:
  - name: c00
    image: ubuntu
    command: ["/bin/bash", "-c", "while true; do echo Rohit Surwade; sleep 5; done"]
  - name: c01
    image: ubuntu
    command: ["/bin/bash", "-c", "while true; do echo DevOps Engineer; sleep 5; done"]
  - name: third-container
    image: centos
    command: ["/bin/bash", "-c", "while true; do echo Actor; sleep 5; done"]
```

10.

****Manifest3****

(This manifest defines a Kubernetes Pod named testpod3 that runs three containers:

The first container (c00) runs an Ubuntu image and prints "Rohit Surwade" every 5 seconds.

The second container (c01) also runs an Ubuntu image and prints "DevOps Engineer" every 5 seconds.

The third container (third_container) runs a CentOS image and prints "Actor" every 5 seconds.

Each container runs its own command in an infinite loop, which is useful for testing or demonstration purposes.

The Pod will continue to run as long as the containers are active. If any of the containers exit, the Pod will not restart them unless a restart policy is specified (the default is Always, but it is not explicitly set in this manifest).)

MULTI CONTAINER POD ENVIRONMENT

```
kind: Pod
apiVersion: v1
metadata:
  name: testpod3
spec:
  containers:
  - name: c00
    image: ubuntu
    command: ["/bin/bash", "-c", "while true; do echo Rohit Surwade; sleep 5; done"]
  - name: c01
    image: ubuntu
    command: ["/bin/bash", "-c", "while true; do echo DevOps Engineer; sleep 5; done"]
  - name: third_container
    image: centos
    command: ["/bin/bash", "-c", "while true; do echo Actor; sleep 5; done"]
```


To enter into a container

```

root@ip-172-31-36-215:/home/ubuntu# kubectl exec testpod3 -c third-container --hostname -i
error: unknown flag: --hostname
See 'kubectl exec --help' for usage.
root@ip-172-31-36-215:/home/ubuntu# kubectl exec testpod3 -c third-container -- hostname -i
10.244.0.6
root@ip-172-31-36-215:/home/ubuntu# kubectl exec testpod3 -c c02 -- hostname -i
Error from server (BadRequest): container c02 is not valid for pod testpod3
root@ip-172-31-36-215:/home/ubuntu# kubectl exec testpod3 -c c01 -- hostname -i
10.244.0.6
root@ip-172-31-36-215:/home/ubuntu# kubectl exec testpod3 -c c00 -- hostname -i
10.244.0.6
root@ip-172-31-36-215:/home/ubuntu# kubectl exec testpod3 -it -c c00 -- bin/bash
root@testpod3:/# ps
  PID TTY          TIME CMD
  131 pts/0    00:00:00 bash
  141 pts/0    00:00:00 ps
root@testpod3:/# ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0 17:19 ?        00:00:00 /bin/bash -c while true; do echo Rohit Surwade; sleep 5; done
root        131        0  0 17:29 pts/0    00:00:00 bin/bash
root        140        1  0 17:29 ?        00:00:00 sleep 5
root        142       131  0 17:29 pts/0    00:00:00 ps -ef
root@testpod3:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run /sbin  srv  sys  usr  var
root@testpod3:/# exit
exit
root@ip-172-31-36-215:/home/ubuntu# kubectl exec testpod3 -it -c c01 -- bin/bash
root@testpod3:/# ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0 17:19 ?        00:00:00 /bin/bash -c while true; do echo DevOps Engineer; sleep 5; done
root        144        1  0 17:30 ?        00:00:00 sleep 5
root        145        0  0 17:30 pts/0    00:00:00 bin/bash
root        153       145  0 17:30 pts/0    00:00:00 ps -ef
root@testpod3:/#

```

15.

```
kubectl exec pod_name -it -c container_name --bin/bash
```